

Q1 Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

A food delivery application calculates the final bill for a customer's order by considering the food cost, delivery charges, and coupons and discounts.

Create two interfaces:

DeliveryCharge with a method calculateDeliveryCharge().

CouponDiscount with a method calculateDiscount().

Create a class FoodOrder that implements both interfaces.

The delivery charge is calculated as follows:

If the food cost is ₹500 or more, the delivery charge is ₹0.

Otherwise, the delivery charge is ₹50.

The coupon discount is calculated as follows:

If the coupon code is SAVE10, the customer receives a 10% discount on the food cost.

Otherwise, no discount is applied.

The final bill is calculated using the formula:

Final Bill = Food Cost + Delivery Charge – Discount

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

Coupon Code

Output Format

Food Cost : <food_cost>

Delivery Charge : <delivery_charge>

Discount : <discount>

Final Bill : <final_bill>

Sample Input

600

SAVE10

Sample Output

Food Cost : 600.0

Delivery Charge : 0.0

Discount : 60.0

Final Bill : 540.0

Your Code

```
import java.util.Scanner;  
interface DeliveryCharge{  
    public void calculateDeliveryCharge();  
}
```

Input

600.0

SAVE10



```
1 import java.util.Scanner;
2 interface DeliveryCharge{
3     public void calculateDeliveryCharge();
4 }
5 interface CouponDiscount{
6     public void calculateDiscount();
7 }
8 public class FoodOrder implements DeliveryCharge,CouponDiscount{
9     public double delivery_charge;
10    public String coupon_code;
11    public double discount;
12    public double food_cost;
13    public double final_bill;
14    public void calculateDeliveryCharge(){
15        if(food_cost>=500){
16            delivery_charge=0;
17        }
18        else{
19            delivery_charge=50;
20        }
21    }
22    public void calculateDiscount(){
23        if(coupon_code.equals("SAVE10")){
24            discount=food_cost*0.1;
25        }
26        else{
27            discount=0;
28        }
29    }
```

600.0

SAVE10

Output

```
Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks


```
25     }
26     else{
27         discount=0;
28     }
29 }
30 public void calculateFinalBill(){
31     final_bill=food_cost+delivery_charge-discount;
32 }
33 public void display(){
34     System.out.println("Food Cost : "+ food_cost);
35     System.out.println("Delivery Charge : "+ delivery_charge);
36     System.out.println("Discount : "+ discount);
37     System.out.println("Final Bill : "+ final_bill);
38 }
39 public static void main(String args[]){
40     Scanner sc = new Scanner(System.in);
41     FoodOrder obj = new FoodOrder();
42     obj.food_cost=sc.nextDouble();
43     sc.nextLine();
44     obj.coupon_code=sc.nextLine();
45     obj.calculateDeliveryCharge();
46     obj.calculateDiscount();
47     obj.calculateFinalBill();
48     obj.display();
49     sc.close();
50 }
51 }
```


Q2 Hotel Billing System (Multiple Inheritance using One Class and One Interface)

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Problem Statement

A hotel wants to generate the final bill for its guests by adding a service charge to the room rent.

Create a class Room that stores the following details:

Room Number

Guest Name

Room Rent

Create an interface ServiceCharge with a method:

`double calculateServiceCharge();`

Create a class HotelBill that extends the Room class and implements the ServiceCharge interface.

The service charge is calculated as follows:

If the room rent is ₹5000 or more, the service charge is 12% of the room rent.

Otherwise, the service charge is 8% of the room rent.

The final bill is calculated as:

Final Bill = Room Rent + Service Charge

Display the room details, room rent, service charge, and final bill.

Input Format

Room Number

Guest Name

Room Rent

Output Format

Room Number : <room_number>

Guest Name : <guest_name>

Room Rent : <room_rent>

Service Charge : <service_charge>

Final Bill : <final_bill>

Sample Input

205

Ananya

6000

Sample Output

Room Number : 205

Guest Name : Ananya

Room Rent : 6000.0

Service Charge : 720.0

Final Bill : 6720.0

Your Code

```
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner in = new Scanner(System.in);
```

Input

205
Ananya
6000


```
2 class Main{
3     public static void main(String args[]){
4         Scanner in = new Scanner(System.in);
5         HotelBill h =new HotelBill();
6         h.Room_number=in.nextInt();
7         in.nextLine();
8         h.Name=in.nextLine();
9         h.Rent=in.nextDouble();
10        h.calculateServiceCharge();
11        h.finalbill();
12        h.display();
13    }
14 }
15 class Room{
16     public int Room_number;
17     public String Name;
18     public double Rent;
19 }
20 interface Service_Charge{
21     public double calculateServiceCharge();
22 }
23 class HotelBill extends Room implements Service_Charge{
24     public double final_bill;
25     public double ServiceCharge;
26     public double calculateServiceCharge(){
27         if(Rent>=5000)
28             ServiceCharge=Rent*0.12;
29         else
30             ServiceCharge=Rent*0.08;
```

Ananya
6000

Output

```
Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks


```
17     public String Name;  
18     public double Rent;  
19 }  
20 interface Service_Charge{  
21     public double calculateServiceCharge();  
22 }  
23 class HotelBill extends Room implements Service_Charge(  
24     public double final_bill;  
25     public double ServiceCharge;  
26     public double calculateServiceCharge(){  
27         if(Rent>=5000)  
28             ServiceCharge=Rent*0.12;  
29         else  
30             ServiceCharge=Rent*0.01;  
31         return ServiceCharge;  
32     }  
33     public void finalbill(){  
34         final_bill=Rent+ServiceCharge;  
35     }  
36     public void display(){  
37         System.out.println("Room Number : "+Room_number);  
38         System.out.println("Guest Name : "+Name);  
39         System.out.println("Room Rent : " +Rent);  
40         System.out.println("Service Charge : "+ServiceCharge);  
41         System.out.println("Final Bill : "+final_bill);  
42     }  
43 }
```

Test Case 2

Test Case 3

10.00 / 10 marks